

Conclusions: NN Classifiers and the Levels of Chess Meaning Integration in the Solver

- 1. The numerical presentation of nuclear knowledge has been overcome.** Chess meanings in the Solver were authored against bare integer encodings (`FigureType = 6` for the king, `FigureColor = 1` for white) that lived only in the expert's head. With the introduced `IS / NOT IS` operators, nucleus conditions are now expressed in the class names of the neural classifiers themselves (`IS king`, `IS white`, `IS knight`, `bishop` for a minor piece). This removes the three problems of the numeric surface: the opacity of the encoding, the coupling of authored meaning to an invisible integer layout, and the silent meaning flip after a classifier is retrained with reordered classes.
- 2. NN classifiers cover the complete nuclear level of chess.** All four chess nucleus abstracts can be backed by trained models shipped in the same `Solver_train` folder: *figure type* (6-class CNN, 1.00 validation accuracy), *figure color* (2-class CNN, 1.00), and the *x/y* coordinates (axis models of the prototype family; deterministic after the 8×8 split, integrable as small categorical vocabularies). Occupancy — the Dummy/empty value `0` of the 2016 convention — is covered today by an occupancy heuristic and, going forward, by the prototype family's explicit `none` class.
- 3. Integration required no change to the reasoning engine.** A classifier plugs in by dropping a folder with a `manifest.json` into the classifier drop folder; the `VocabularyRegistry` turns its ordered `classNames` into a name ↔ integer vocabulary, and `RuleIs` translates names to integers at rule-construction time. The GA matching, conjunction, inheritance and persistence continue to operate on integers — the NN vocabulary is a surface adapter, not an engine rewrite, and new classifiers (traffic signs, terrain classes, any categorical nucleus) plug in without code changes.
- 4. The higher levels of meaning complexity inherit the symbolic surface for free.** Because classifiers bind only at the nuclear level, every level of the meaning hierarchy builds on them unchanged: *primitive* abstracts strengthen by names (`King = FigureType IS king`); *AR1* abstracts compose classifier-backed nuclei into positional units (`Figure`); *composite* abstracts and their virtual/polymorphic specifications express relations over them (`FieldUnderCheck`); *set* and *action* abstracts, goals and plans consume these without ever seeing a class name. The division of labor is preserved: NNs answer only nuclear-level questions, while relations and strategy remain symbolic.
- 5. The perception loop is closed and verified.** A board image is converted into a Solver Situation through the same classifiers: per-cell type/color classification → name-to-integer mapping → IdGroups of nucleus instances → GA matching, in the same shape as the FEN and drag-and-drop paths. The pipeline was demonstrated end-to-end, reconstructing the full start position (all 32 pieces, correct FEN) from a board image and passing the resulting Situation through the meaning base.

6. **A fifth, relational level of classification is prepared.** The Siamese neighborhood detector (`neighbor / not_neighbor` , 0.839 validation accuracy) is trained and registered as a `relation - kind` classifier. It does not bind to a nucleus — a relation judges two instances, not one value — and marks the next integration surface: perception-side grid verification in the short term, and NN-backed relational predicates at the AR1/composite level as future work.
7. **Retraining drift is now detectable instead of silent.** Saved rules persist the symbolic form (`{oper: IS, value: "king", classifier: "type_classifier"}`) together with the classifier identity, and the manifest remains authoritative for the current index mapping; a reordered retrain therefore produces a resolvable mismatch rather than a silently inverted meaning. Legacy numeric abstracts are rescued by unique-vocabulary inference at load time.
8. **Remaining steps are refinements, not redesign.** The perception bridge should resolve names through the vocabulary registry instead of a hardcoded map; occupancy should become a learned class; the recognized Situation should be rendered back onto the board interface; the `IS` surface should be extended to the composite editor; and the per-request Python subprocess should become a persistent inference sidecar. None of these touches the core result: numeric nuclear presentations have been replaced by the classifiers' own vocabularies, at every level of the chess meaning hierarchy, with the reasoning engine untouched.